Author: Rick
Date: 2026-10-28
To: Clio Vega (cc Robin Langer)
Title: FPSAC 2027 cut: LG/OG negative-degree Schubert integrals — ready for review
Work-in-progress commit: `6dbecb8`
PDF at that commit: `https://github.com/grandpa-rick/work-in-progress/blob/6dbecb8/papers/lg-og-fpsac-cut/main.pdf`
Source: `https://github.com/grandpa-rick/work-in-progress/blob/6dbecb8/papers/lg-og-fpsac-cut/main.tex`
Status: NOT submitted. Use 6dbecb8, not the earlier bac1058.

Time-critical

The FPSAC 2027 deadline is **2026-11-15**. Nothing is submitted without Robin's yes. Robin owns the venue yes/no and the six blue `\robin{}` fields: authorship/presenter, affiliation, human acknowledgements, AI-declaration wording, the model list, and the human role.

What the cut contains

- 12 counted pages including the bibliography; the AI declaration follows on two uncounted pages. 0 TODOs.
- The LG main line uses only (G1) and (LS): the Demazure character formula and Littelmann's path model. This covers Thm 1.1 (strict reciprocity), the recursion ($\star$) (Prop 4.3, proved by peel + polynomiality), the integer window, the half-integer law (Thm 1.5) and Cor 1.6.
- Buch–Ravikumar (BR) is cited in exactly two remarks. Rem 4.4 gives the Pieri values $c_\rho = (-1)^{|\theta|-1}V(\theta)$ as the $p = 1$ case of their Cor 5.6, with no novelty claim. Rem 4.5 gives the OG K-theoretic sketch via their Lemma 4.5. The OG theorem itself is cited as classical.
- Check run: a fresh script re-derives every Table 1 value and every displayed formula from the printed statements. It found 0 failures (`scripts/prove260/pdf_check.py`).
- Cold referee read (PROVE 260b): 0 mathematical errors. It fixed three precision points. The LS bijection now defines $a_s = 1$. The $\wedge U$ Euler-characteristic identity needs (G1) for χ at all weights, not only dominant ones (the paper doesn't use it). The $(3, 2, 1)$ value at $-7/2$ is now described as attaining the Thm 1.5(iii) bound, because it is not the equality case.

Requests for Clio

1. Line-read §3 (LS proof) and the proof of Prop 4.3. Nobody but me has read these.
2. Line-read the compressed proof of Thm 1.5 (HZ) in §6. It is the most compressed part.
3. Check that the AI declaration describes your reviews accurately: Thm Z (uid451), the window for both types (uid457), the $m \leq 6$ census, and the 1243-shape Pieri check.

The cut is attached as `main.pdf`, together with this note.